Reactivity of Halogenoalkanes (Cambridge (CIE) A Level Chemistry): Revision Note

Reactivity of Halogenoalkanes

- The halogenoalkanes have different **rates** of **substitution reactions**
- Since substitution reactions involve **breaking** the **carbon-halogen** bond the **bond energies** can be used to explain their different reactivities

Halogenoalkane bond Eenergy table

Bond	Bond Energy / kJ mol^{-1}
C–F	467 (strongest bond)
C–Cl	346
C–Br	290
C–I	228 (weakest bond)

- The table above shows that the C-I bond requires the least energy to break, and is therefore the **weakest** carbon-halogen bond
- During substitution reactions the C-I bond will therefore **heterolytically** break as follows:

$$R_3C\text{-}I + OH^- \rightarrow R_3C\text{-}OH + I^-$$

halogenoalkane → alcohol

- The C-F bond, on the other hand, requires the most energy to break and is, therefore, the **strongest** carbon-halogen bond
- Fluoroalkanes will therefore be less likely to undergo substitution reactions

Aqueous silver nitrate

- Reacting halogenoalkanes with **aqueous silver nitrate solution** will result in the formation of a **precipitate**
- The **rate of formation** of these precipitates can also be used to determine the reactivity of the halogenoalkanes

Halogenoalkane precipitates

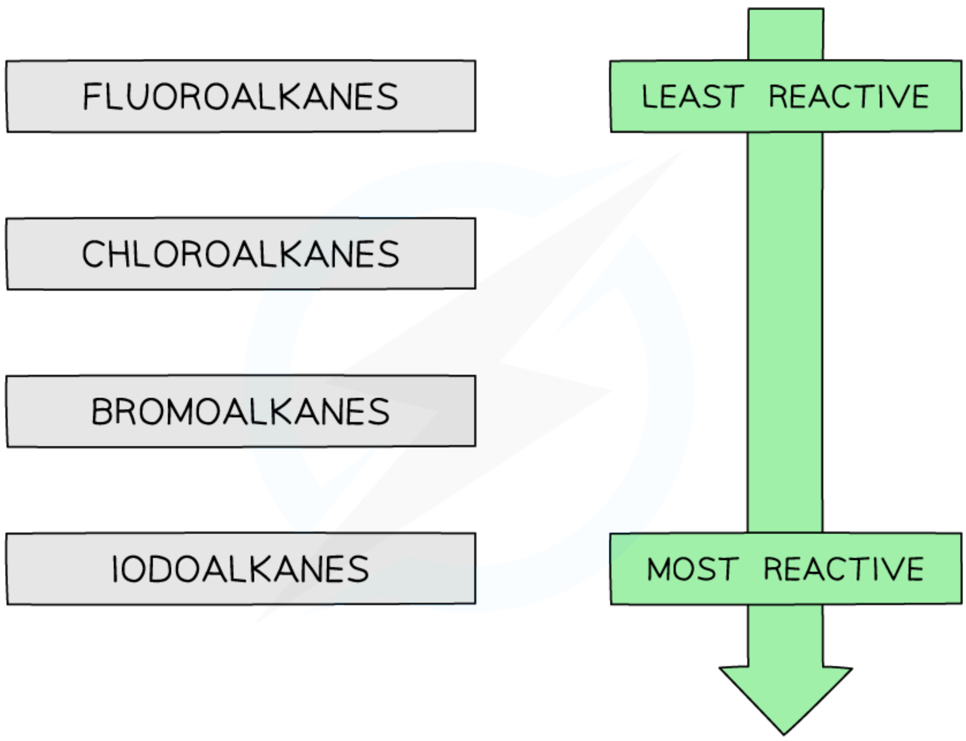
- Chlorides
  - White (silver chloride) precipitate

- Bromides
  - Cream (silver bromide)

- Iodides
  - Yellow (silver iodide)

- The formation of the pale yellow silver iodide is the fastest (fastest **nucleophilic substitution** reaction)
- The formation of the silver fluoride is the slowest (slowest **nucleophilic substitution** reaction)
- This confirms that fluoroalkanes are the least reactive and iodoalkanes are the most reactive halogenoalkanes

The trend in reactivity of halogenoalkanes



The halogenoalkanes become more reactive as you move down the group